COMP7024 Programming for Data Science

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```r
orders <- read_csv("project_orders.csv")
products <- read_csv("project_product_info.csv")

head(orders)
```

```
## # A tibble: 6 x 7
##   InvoiceNo StockCode Quantity InvoiceDate    UnitPrice CustomerID Country
##   <chr>     <chr>        <dbl> <chr>              <dbl>      <dbl> <chr>
## 1 536365    85123A           6 1/12/2010 8:26      2.55      17850 United Kingd~
## 2 536365    71053            6 1/12/2010 8:26      3.39      17850 United Kingd~
## 3 536365    84406B           8 1/12/2010 8:26      2.75      17850 United Kingd~
## 4 536365    84029G           6 1/12/2010 8:26      3.39      17850 United Kingd~
## 5 536365    84029E           6 1/12/2010 8:26      3.39      17850 United Kingd~
## 6 536365    22752            2 1/12/2010 8:26      7.65      17850 United Kingd~
```

```r
head(products)
```

```
## # A tibble: 6 x 2
##   StockCode Description
##   <chr>     <chr>
## 1 85123A    WHITE HANGING HEART T-LIGHT HOLDER
## 2 71053     WHITE METAL LANTERN
## 3 84406B    CREAM CUPID HEARTS COAT HANGER
## 4 84029G    KNITTED UNION FLAG HOT WATER BOTTLE
## 5 84029E    RED WOOLLY HOTTIE WHITE HEART.
## 6 22752     SET 7 BABUSHKA NESTING BOXES
```

Question 1

Explanation

The objective of carrying out this analysis is to assess the performance of the products by evaluating the total revenue generated by each product. This is achieved by using the orders data set, which contains information regarding each transaction in terms of the quantity of each product sold, the price at which it is being sold, etc. Additionally, the product data set contains descriptive information regarding each product. This information is linked together by matching the StockCode variable in each data set.

Before carrying out the analysis, it is important to clean the data to ensure that it is accurate. This is achieved by dropping rows that contain cancelled orders, as identified by the invoice column starting with the letter "C." Additionally, rows with negative quantities in the orders data set are removed. This is because negative values in the quantity column indicate invalid sales.

Revenue generated by each transaction is calculated by multiplying the quantity of each product by the price at which it is being sold. This is achieved by the formula Revenue = Quantity * Unit Price.

After that, the data is grouped by each product to calculate the total revenue generated by each product. This is achieved by ranking the products from highest to lowest revenue generated. Additionally, the percentage contribution of the top three products to the total revenue is calculated to assess the percentage contribution of each product to the total revenue generated by the organization.

Code

```r
# Data Cleaning
orders_clean <- orders %>%
  filter(!str_detect(InvoiceNo, "^C")) %>%   # remove cancelled orders
  filter(Quantity > 0) %>%                   # remove negative quantities
  mutate(StockCode = toupper(StockCode))    # standardize stock codes

products <- products %>%
  mutate(StockCode = toupper(StockCode))

# Join datasets
data <- orders_clean %>%
  left_join(products, by = "StockCode")

# Remove unwanted descriptions
data <- data %>%
  filter(!Description %in% c("Adjust bad debt", "adjustment", "thrown away")) %>%
  filter(!is.na(Description))

# Calculate revenue
data <- data %>%
  mutate(Revenue = Quantity * UnitPrice)

# Revenue by product
product_revenue <- data %>%
  group_by(StockCode, Description) %>%
  summarise(TotalRevenue = sum(Revenue), .groups = "drop") %>%
  arrange(desc(TotalRevenue))

# Top 10 products
```

```r
top10 <- product_revenue %>%
  slice_max(TotalRevenue, n = 10)

# Bottom 10 products
bottom10 <- product_revenue %>%
  slice_min(TotalRevenue, n = 10)

# Top 3 contribution
top3 <- product_revenue %>%
  slice_max(TotalRevenue, n = 3)

total_revenue <- sum(product_revenue$TotalRevenue)

top3_percentage <- sum(top3$TotalRevenue) / total_revenue * 100

# Display tables
kable(top10, caption = "Top 10 Products by Revenue")
```

Table 1: Top 10 Products by Revenue

StockCode	Description	TotalRevenue
85123A	WHITE HANGING HEART T-LIGHT HOLDER	212942.56
DOT	DOTCOM POSTAGE	206248.77
22423	REGENCY CAKESTAND 3 TIER	174484.74
23843	PAPER CRAFT , LITTLE BIRDIE	168469.60
M	Manual	156225.64
47566	PARTY BUNTING	99504.33
85099B	JUMBO BAG RED RETROSPOT	94340.05
23166	MEDIUM CERAMIC TOP STORAGE JAR	81700.92
POST	POSTAGE	78101.88
23084	RABBIT NIGHT LIGHT	66964.99

```r
kable(bottom10, caption = "Bottom 10 Products by Revenue")
```

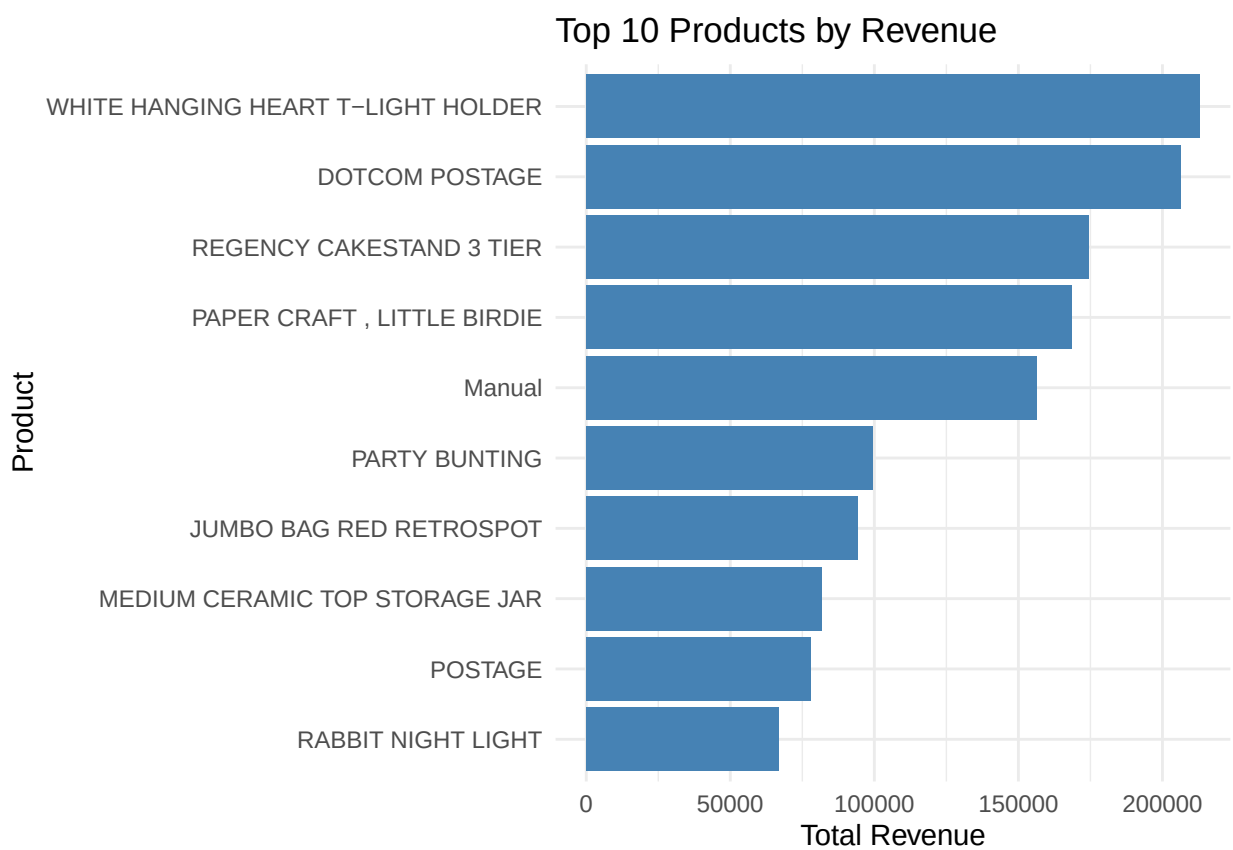
Table 2: Bottom 10 Products by Revenue

StockCode	Description	TotalRevenue
23001	TRAVEL CARD WALLET DOTCOMGIFTSHOP	0.000
35600A	Found by jackie	0.000
PADS	PADS TO MATCH ALL CUSHIONS	0.003
21268	VINTAGE BLUE TINSEL REEL	0.840
90084	PINK CRYSTAL GUITAR PHONE CHARM	0.850
84201C	HAPPY BIRTHDAY CARD TEDDY/CAKE	0.950
84206B	CAT WITH SUNGLASSES BLANK CARD	0.950
21009	ETCHED GLASS STAR TREE DECORATION	1.250
35597A	DUSTY PINK CHRISTMAS TREE 30CM	1.250
35597B	BLACKCHRISTMAS TREE 30CM	1.250
84569C	PACK 4 FLOWER/BUTTERFLY PATCHES	1.250
84743C	ORANGE FELT VASE + FLOWERS	1.250

```r
top3_percentage
```

```
## [1] 5.305872
```

```r
#Visualisation
ggplot(top10, aes(x=reorder(Description, TotalRevenue), y=TotalRevenue)) +
  geom_bar(stat="identity", fill="steelblue") +
  coord_flip() +
  labs(
    title="Top 10 Products by Revenue",
    x="Product",
    y="Total Revenue"
  ) +
  theme_minimal()
```



Interpretation

The results show that there are products that yield significantly higher revenues than others. For instance, products like WHITE HANGING HEART T-LIGHT HOLDER and REGENCY CAKESTAND 3 TIER are among those that yield high revenues. This suggests that there is a small group of products that yield a significant amount of revenue. The revenue distribution is a normal occurrence in retail businesses, in which a small group of products is responsible for generating most of the revenue. From the graph, it is clear that a small percentage of products contribute significantly higher revenue compared to the rest. WHITE HANGING HEART T-LIGHT HOLDER generates the highest revenue, followed by DOTCOM POSTAGE

and then REGENCY CAKESTAND 3 TIER. This high difference between the top-generating products and the rest implies that revenue is generated by a small percentage of popular products that probably enjoy high and constant demand. However, there is also a group of products that yield low or even no revenue at all. This could be due to low sales of these products, or it could be that these products are simply entries.

From a business point of view, these findings could be useful in: - Identifying products that yield high revenue, which could be given priority in terms of promotional strategies - Identifying products that yield low revenue, which could be given promotional strategies - Identifying how revenue is being distributed among products in general.

Question 2

Explanation

The purpose of this study is to find the most valuable customers. The purchasing behavior of these customers is also analyzed. The customer value is calculated by determining the total revenue obtained from the transactions of these customers. First, the transactions without a CustomerID are removed, as these cannot be attributed to any customer. Then, the total expenditure of each customer is calculated by adding up all the revenue obtained in their transactions. Once the total expenditure of each customer is calculated, these customers are ranked in order to find the top five customers who spend the most. These customers form the most valuable customer group of the company. Further analysis of the purchasing behavior of these customers is done by determining the products that these customers purchase the most. This is done by calculating the total quantity of these products purchased by these customers. This method of analysis provides knowledge about the purchasing behavior of the most important customers of the company.

Calculate Customer Spending

```r
# Step 1: Remove rows with missing CustomerID
data_clean <- data %>%
  filter(!is.na(CustomerID))

# Step 2: Calculate total spending per customer
customer_spending <- data_clean %>%
  group_by(CustomerID) %>%
  summarise(
    TotalSpending = sum(Revenue, na.rm = TRUE)
  ) %>%
  arrange(desc(TotalSpending))
```

```r
class(data)
```

```
## [1] "tbl_df"     "tbl"        "data.frame"
```

```r
ls()
```

```
##  [1] "bottom10"          "customer_spending" "data"
##  [4] "data_clean"        "orders"            "orders_clean"
##  [7] "product_revenue"   "products"          "top10"
## [10] "top3"              "top3_percentage"   "total_revenue"
```

```r
ls()
```

```
##  [1] "bottom10"          "customer_spending" "data"
##  [4] "data_clean"        "orders"            "orders_clean"
##  [7] "product_revenue"   "products"          "top10"
## [10] "top3"              "top3_percentage"   "total_revenue"
```

Identify Top 5 Highest Spending Customer

```r
top_customers <- customer_spending %>%
  slice_head(n = 5)

# Display result
knitr::kable(top_customers, caption = "Top 5 Highest Spending Customers")
```

Table 3: Top 5 Highest Spending Customers

CustomerID	TotalSpending
14646	287627.1
18102	265212.8
17450	208144.5
16446	168472.5
14911	153055.5

Analyse Purchasing Behaviour of Top Customers

```r
top_customer_products <- data_clean %>%
  filter(CustomerID %in% top_customers$CustomerID) %>%
  group_by(CustomerID, Description) %>%
  summarise(
    TotalQuantity = sum(Quantity, na.rm = TRUE),
    .groups = "drop"
  )
```

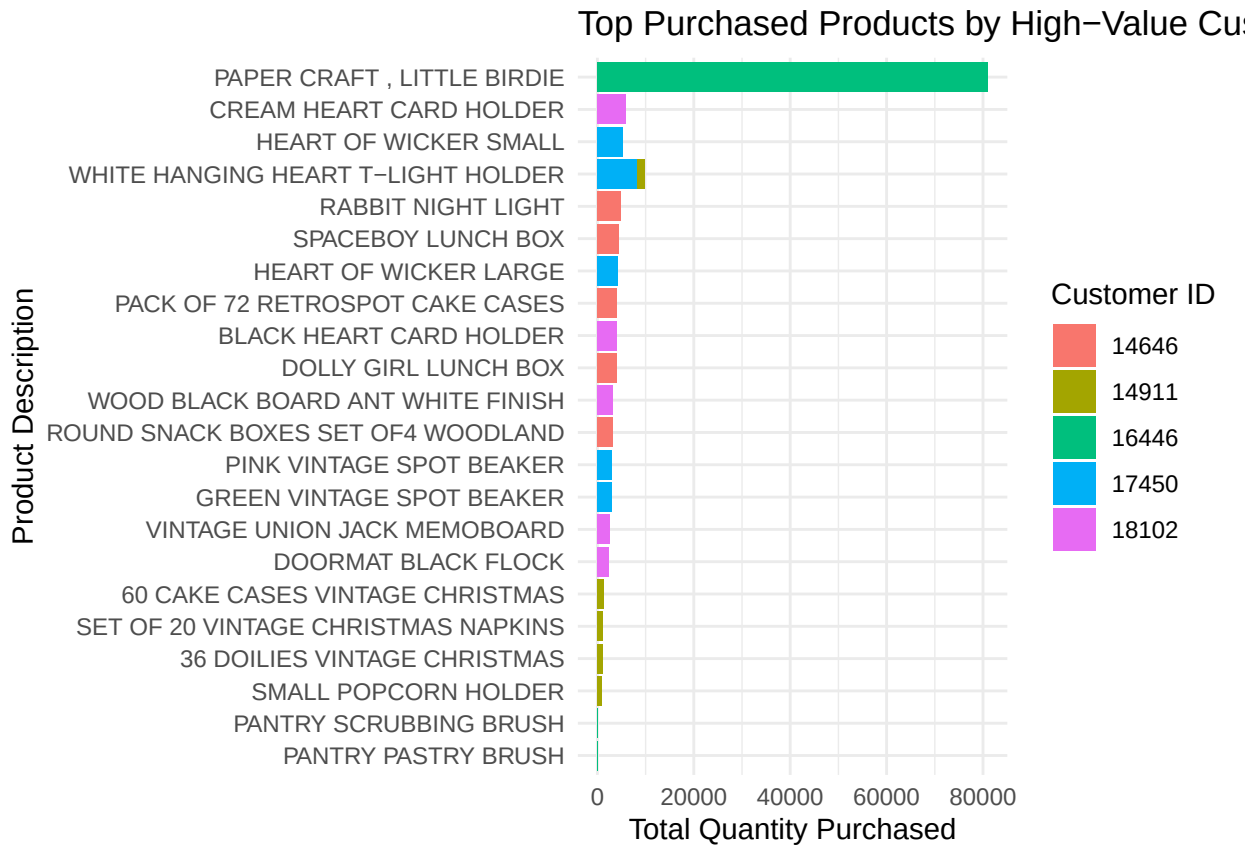
Identify top products purchased by each customer

```r
top_products_per_customer <- top_customer_products %>%
  group_by(CustomerID) %>%
  slice_max(TotalQuantity, n = 5) %>%
  arrange(CustomerID, desc(TotalQuantity))

# Display table
knitr::kable(top_products_per_customer,
             caption = "Top Purchased Products by High Value Customers")
```

Table 4: Top Purchased Products by High Value Customers

CustomerID	Description	TotalQuantity
14646	RABBIT NIGHT LIGHT	4801
14646	SPACEBOY LUNCH BOX	4492
14646	PACK OF 72 RETROSPOT CAKE CASES	4104
14646	DOLLY GIRL LUNCH BOX	4096
14646	ROUND SNACK BOXES SET OF4 WOODLAND	3120
14911	WHITE HANGING HEART T-LIGHT HOLDER	1540
14911	60 CAKE CASES VINTAGE CHRISTMAS	1440
14911	SET OF 20 VINTAGE CHRISTMAS NAPKINS	1140
14911	36 DOILIES VINTAGE CHRISTMAS	1068
14911	SMALL POPCORN HOLDER	928
16446	PAPER CRAFT , LITTLE BIRDIE	80995
16446	PANTRY PASTRY BRUSH	1
16446	PANTRY SCRUBBING BRUSH	1
17450	WHITE HANGING HEART T-LIGHT HOLDER	8228
17450	HEART OF WICKER SMALL	5286
17450	HEART OF WICKER LARGE	4282
17450	GREEN VINTAGE SPOT BEAKER	3056
17450	PINK VINTAGE SPOT BEAKER	3056
18102	CREAM HEART CARD HOLDER	5946
18102	BLACK HEART CARD HOLDER	4104
18102	WOOD BLACK BOARD ANT WHITE FINISH	3205
18102	VINTAGE UNION JACK MEMOBOARD	2600
18102	DOORMAT BLACK FLOCK	2426

```r
#Visualization

top_customer_products <- data_clean %>%
  filter(CustomerID %in% top_customers$CustomerID) %>%
  group_by(CustomerID, Description) %>%
  summarise(TotalQuantity = sum(Quantity), .groups="drop")

top_products_per_customer <- top_customer_products %>%
  group_by(CustomerID) %>%
  slice_max(TotalQuantity, n = 5)
library(ggplot2)

ggplot(top_products_per_customer,
       aes(x = reorder(Description, TotalQuantity),
           y = TotalQuantity,
           fill = factor(CustomerID))) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(
    title = "Top Purchased Products by High-Value Customers",
    x = "Product Description",
    y = "Total Quantity Purchased",
    fill = "Customer ID"
  ) +
  theme_minimal()
```




```r
# Check dataset structure
str(data_clean)
```

```
## tibble [412,426 x 9] (S3: tbl_df/tbl/data.frame)
##  $ InvoiceNo  : chr [1:412426] "536365" "536365" "536365" "536365" ...
##  $ StockCode  : chr [1:412426] "85123A" "85123A" "71053" "84406B" ...
##  $ Quantity   : num [1:412426] 6 6 6 8 6 6 2 6 6 6 ...
##  $ InvoiceDate: chr [1:412426] "1/12/2010 8:26" "1/12/2010 8:26" "1/12/2010 8:26" "1/12/2010 8:26" .
##  $ UnitPrice  : num [1:412426] 2.55 2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 ...
##  $ CustomerID : num [1:412426] 17850 17850 17850 17850 17850 ...
##  $ Country    : chr [1:412426] "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" .
##  $ Description: chr [1:412426] "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE HANGING HEART T-LIGHT HOL
##  $ Revenue    : num [1:412426] 15.3 15.3 20.3 22 20.3 ...
```

```r
# Check number of unique customers
n_distinct(data_clean$CustomerID)
```

```
## [1] 4339
```

```r
# Verify top customers
print(top_customers)
```

```
## # A tibble: 5 x 2
##   CustomerID TotalSpending
```

```
##        <dbl>         <dbl>
## 1      14646       287627.
## 2      18102       265213.
## 3      17450       208144.
## 4      16446       168472.
## 5      14911       153056.
```

Interpretation

The results show that a small number of customers generate a significant portion of the company's revenue. The top five customers have spent considerably more than the average customer, indicating that they are highly valuable to the business. By analysing their purchasing behaviour, it becomes clear that these customers frequently purchase specific products in large quantities. This suggests strong product preferences among high-value customers. Further, from the above bar chart, the products most commonly bought by the five highest-spending customers, based on the overall quantities bought, are shown. From the above chart, it is evident that the high-value customers are buying certain products in significantly higher quantities than the others. For example, PAPER CRAFT, LITTLE BIRDIE is bought in significantly higher quantities by the high-value customers compared to the quantities of the other products bought by them.

As can be understood from the above chart, the high-value customers may be bulk buyers or retailers who are looking to buy these products for resale purposes rather than for consumption purposes by the customers themselves, as they are buying the products in bulk quantities and not in smaller quantities for consumption purposes.

The fact that certain products are being bought by all the high-value customers is a clear indication of the high demand for these products by the high-value customers.

As can be understood from the above analysis, the high-value customers provide significant insight for the company, and the company can gain from the fact that by catering to the high-value customers, the overall stability of the revenue can be ensured.

Q3

Explanation

The objective of this analysis is to evaluate the geographical distribution of the sales by determining the revenue generated and the demand for the products in different countries. This will provide valuable insights into the locations that generate the highest revenue and the popularity of the products. To do this, the transaction data set is used after eliminating the canceled orders and the invalid transactions. The revenue generated per transaction has already been computed as the product of Quantity and Unit Price. The data set is grouped based on the revenue generated per transaction, and the total revenue generated per country is computed. This will enable the determination of the countries that generate the highest revenue for the firm's total sales. This will help in the evaluation of the geographical market concentration of the firm and determine whether the firm generates its revenue through domestic and international markets. Besides the revenue generated per transaction, the demand for the products is also evaluated by determining the most popular product sold in each country. This is achieved by grouping the data set based on the country and the description of the product, and the total quantity sold is computed. The product with the highest quantity sold is determined as the most popular product sold in that country. The integration of these two evaluations will enable the determination of the macro-level and the micro-level analysis of the firm's sales and the products sold in the different countries.

```r
# Calculate total revenue by country
country_revenue <- data %>%
  group_by(Country) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE)) %>%
  arrange(desc(TotalRevenue))

# Display table
kable(country_revenue, caption = "Total Revenue by Country")
```

Table 5: Total Revenue by Country

Country	TotalRevenue
United Kingdom	9440957.77
EIRE	298532.43
Netherlands	293140.27
Germany	239772.03
France	225434.63
Australia	141021.41
Spain	72565.62
Switzerland	59581.72
Belgium	42169.87
Norway	39265.02
Sweden	38596.50
Portugal	38545.59
Japan	37905.97
Singapore	33651.53
Finland	28471.75
Hong Kong	22515.58
Channel Islands	21434.89
Denmark	19862.15
Italy	18153.16
Cyprus	14791.93
Austria	11109.38
Israel	8168.37
Poland	7642.00
Iceland	4817.30
Unspecified	4800.55
Greece	4795.62
USA	3778.84
Canada	3695.10
Malta	2926.49
United Arab Emirates	1967.68
Lithuania	1780.46
Lebanon	1711.58
European Community	1459.65
Brazil	1197.15
RSA	1021.78
Czech Republic	826.74
Bahrain	824.94
Saudi Arabia	145.92

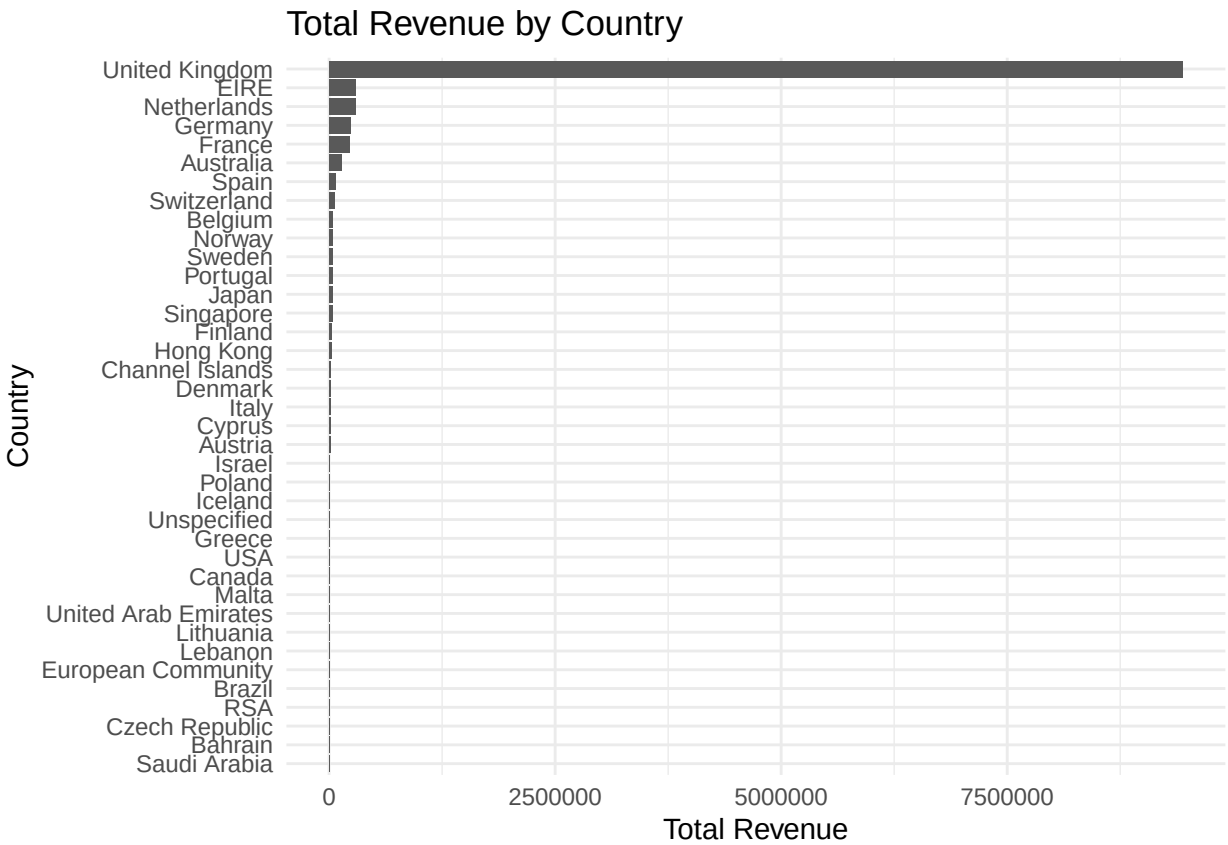
```r
# Calculate quantity sold for each product in each country
country_products <- data %>%
  group_by(Country, Description) %>%
  summarise(TotalQuantity = sum(Quantity, na.rm = TRUE), .groups = "drop")

# Identify most sold product in each country
top_product_country <- country_products %>%
  group_by(Country) %>%
  slice_max(TotalQuantity, n = 1)

# Display table
kable(top_product_country, caption = "Most Popular Product by Country")
```

Table 6: Most Popular Product by Country

Country	Description	TotalQuantity
Australia	MINI PAINT SET VINTAGE	2952
Austria	SET 12 KIDS COLOUR CHALK STICKS	288
Bahrain	ICE CREAM SUNDAE LIP GLOSS	96
Belgium	FAIRY CAKES NOTEBOOK A7 SIZE	640
Brazil	DOLLY GIRL LUNCH BOX	24
Brazil	GREEN REGENCY TEACUP AND SAUCER	24
Brazil	PINK REGENCY TEACUP AND SAUCER	24
Brazil	ROSES REGENCY TEACUP AND SAUCER	24
Brazil	SET OF 4 PANTRY JELLY MOULDS	24
Brazil	SET OF 6 SPICE TINS PANTRY DESIGN	24
Brazil	SET/3 RED GINGHAM ROSE STORAGE BOX	24
Brazil	SMALL HEART FLOWERS HOOK	24
Canada	RETRO COFFEE MUGS ASSORTED	504
Channel Islands	RAIN PONCHO	407
Cyprus	HEART DECORATION PAINTED ZINC	384
Czech Republic	WOODEN STAR CHRISTMAS SCANDINAVIAN	72
Czech Republic	WOODEN TREE CHRISTMAS SCANDINAVIAN	72
Denmark	RED HARMONICA IN BOX	288
EIRE	WHITE HANGING HEART T-LIGHT HOLDER	2184
European Community	CHARLIE AND LOLA FIGURES TINS	24
European Community	CHOCOLATE 1 WICK MORRIS BOX CANDLE	24
European Community	RED ROCKING HORSE HAND PAINTED	24
European Community	ROCKING HORSE GREEN CHRISTMAS	24
European Community	ROCKING HORSE RED CHRISTMAS	24
European Community	SET OF 60 PANTRY DESIGN CAKE CASES	24
European Community	SET OF 60 VINTAGE LEAF CAKE CASES	24
European Community	SET/3 VANILLA SCENTED CANDLE IN BOX	24
European Community	WHITE ROCKING HORSE HAND PAINTED	24
Finland	PINK 3 PIECE POLKADOT CUTLERY SET	1104
France	RABBIT NIGHT LIGHT	4024
Germany	ROUND SNACK BOXES SET OF4 WOODLAND	1233
Greece	4 LAVENDER BOTANICAL DINNER CANDLES	48
Greece	4 PEAR BOTANICAL DINNER CANDLES	48
Hong Kong	PINK 3 PIECE POLKADOT CUTLERY SET	240
Iceland	3D DOG PICTURE PLAYING CARDS	252
Israel	WOODLAND CHARLOTTE BAG	130

Country	Description	TotalQuantity
Italy	FEATHER PEN,COAL BLACK	336
Japan	RABBIT NIGHT LIGHT	3408
Lebanon	ASSTD FRUIT+FLOWERS FRIDGE MAGNETS	24
Lithuania	FELTCRAFT DOLL ROSIE	48
Lithuania	IVORY ENCHANTED FOREST PLACEMAT	48
Lithuania	RED HARMONICA IN BOX	48
Malta	GRAND CHOCOLATECANDLE	81
Netherlands	RABBIT NIGHT LIGHT	4801
Norway	SMALL FOLDING SCISSOR(POINTED EDGE)	576
Poland	STRAWBERRY CERAMIC TRINKET BOX	168
Portugal	mailout	376
RSA	mailout	20
Saudi Arabia	ASSORTED BOTTLE TOP MAGNETS	12
Saudi Arabia	HOMEMADE JAM SCENTED CANDLES	12
Saudi Arabia	PLASTERS IN TIN CIRCUS PARADE	12
Saudi Arabia	PLASTERS IN TIN SKULLS	12
Saudi Arabia	PLASTERS IN TIN STRONGMAN	12
Singapore	CHRISTMAS TREE PAINTED ZINC	384
Spain	PINK 3 PIECE POLKADOT CUTLERY SET	2178
Sweden	MINI PAINT SET VINTAGE	2916
Switzerland	PLASTERS IN TIN WOODLAND ANIMALS	639
USA	SET 12 COLOURING PENCILS DOILY	88
United Arab Emirates	ASSORTED CHEESE FRIDGE MAGNETS	72
United Arab Emirates	BIG DOUGHNUT FRIDGE MAGNETS	72
United Kingdom	PAPER CRAFT , LITTLE BIRDIE	80995
Unspecified	WORLD WAR 2 GLIDERS ASSTD DESIGNS	96

```r
#Revenue Comparison Across Countries
ggplot(country_revenue,
       aes(x = reorder(Country, TotalRevenue),
           y = TotalRevenue)) +
  geom_bar(stat = "identity") +
  coord_flip() +
  labs(
    title = "Total Revenue by Country",
    x = "Country",
    y = "Total Revenue"
  ) +
  theme_minimal()
```



Interpretation

The results indicate that the United Kingdom generates the highest total revenue by a wide margin, contributing the majority of total sales. This is expected since the retailer is based in the UK, meaning that it is expected that this market would contain the greatest number of customers with a higher purchasing activity. In addition, a number of other European countries, including EIRE, the Netherlands, Germany, and France, contribute a significant amount of revenue to total sales. This implies that the retailer has managed to expand beyond its domestic market and establish operations within the European region. The fact that these countries perform well implies that this market segment is a key contributor to international operations for this company. In contrast, a number of countries contribute a smaller amount of revenue compared to others. This can be attributed to a number of reasons, including a smaller number of customers or a lack of marketing within these regions.

Analysis of product demand in different countries indicates that different countries have different preferences for different products. Some products may be found to be top sellers in different countries, whereas some may be found to be highly popular in different regions. For instance, some products may be found to have particularly high demand in different regions of Europe. From a business strategy point of view, it is observed that such analyses are highly beneficial for business decision-making. By analyzing high-performing international markets, it is possible for the company to invest in these regions where high demand for their products exists. At the same time, analyzing regional product preferences may be highly beneficial for business in terms of promoting different products.

Overall, it is observed that the analysis highlights the significance of geographical sales patterns, which are highly beneficial for businesses in optimizing their marketing strategies in different regions of the world.

Question 4

Explaination

This study aims to examine the sales trends over time through the use of the InvoiceDate variable. The main purpose of this analysis is to identify the sales trends over the months and understand the performance of products in different months.To start the analysis, the month and year are extracted from the Invoice variable. This allows the sales transactions to be grouped according to the months in which they were made. Consistent with the assumptions provided in the assignment, the sales transactions that were cancelled are not considered in the analysis. The cancelled sales transactions are identified as the ones where the invoice number starts with the letter "C." After the data cleansing process, the total revenue generated in each month is then computed. This allows the identification of the months where the highest sales are generated. Moreover, this analysis allows the identification of the months where the sales are fluctuating. Finally, a time series plot is then created to show the sales trends over the months. This allows the identification of the months where the sales revenue is augmented and diminished. Aside from the sales trends over the months, the performance of the products is also analyzed over the months. The analysis identified the products where the lowest sales revenue is generated over the months.

```r
# Load libraries
library(dplyr)
library(lubridate)
library(ggplot2)
library(knitr)

# Convert date
data <- data %>%
  mutate(InvoiceDate = dmy_hm(InvoiceDate)) %>%
  filter(!is.na(InvoiceDate))

# Remove cancelled invoices
data <- data %>%
  filter(!grepl("^C", InvoiceNo))

# Calculate revenue
data <- data %>%
  mutate(Revenue = Quantity * UnitPrice)

# Create time variables
data <- data %>%
  mutate(
    Year = year(InvoiceDate),
    Month = month(InvoiceDate),
    YearMonth = floor_date(InvoiceDate, "month")
  )
#Monthly Revenue Calculation
monthly_revenue <- data %>%
  group_by(YearMonth) %>%
  summarise(
    TotalRevenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  )

#Highest Sales Months
top_months <- monthly_revenue %>%
```

```r
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5)

kable(monthly_revenue, caption = "Monthly Revenue")
```

Table 7: Monthly Revenue

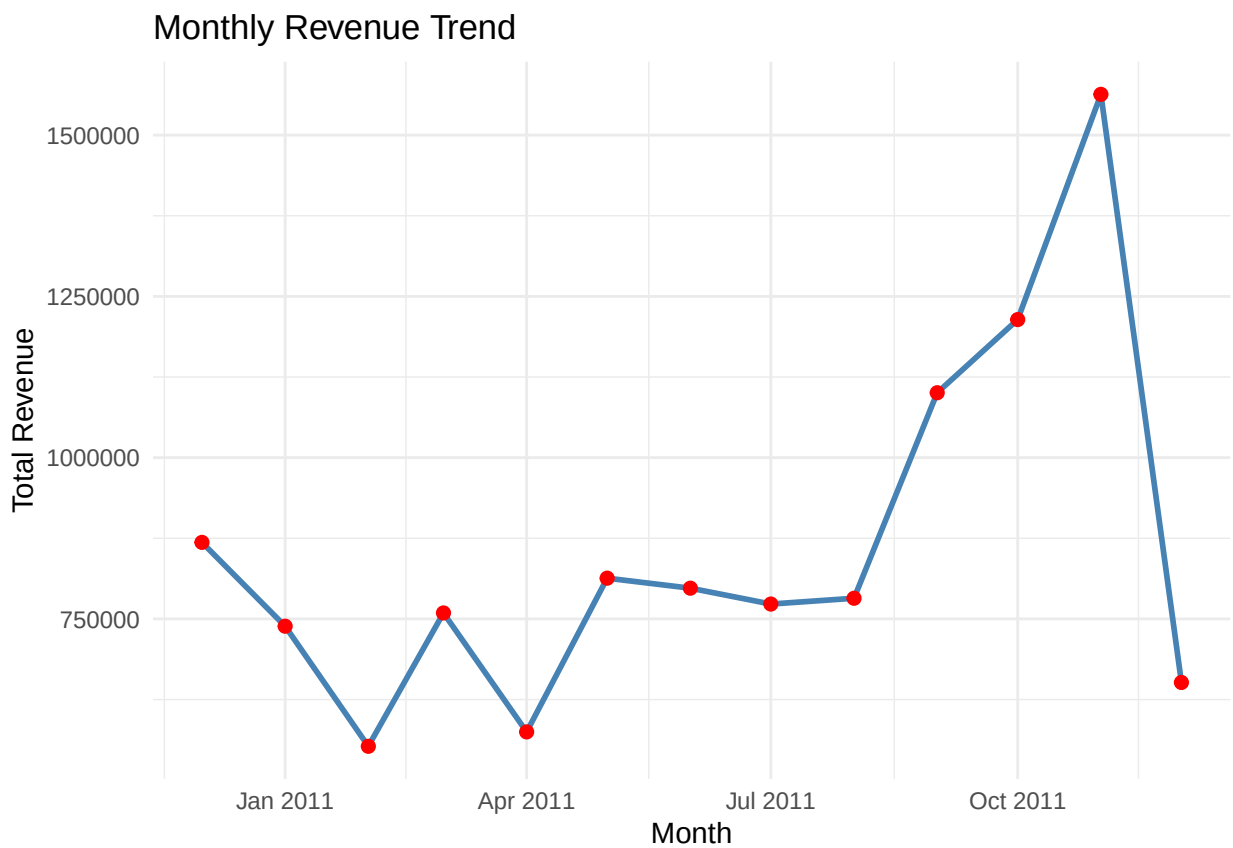
YearMonth	TotalRevenue
2010-12-01	868633.0
2011-01-01	738670.5
2011-02-01	552508.4
2011-03-01	759222.7
2011-04-01	574832.9
2011-05-01	813138.0
2011-06-01	797622.3
2011-07-01	773038.7
2011-08-01	782023.9
2011-09-01	1100624.5
2011-10-01	1214039.9
2011-11-01	1563225.5
2011-12-01	651459.0

```r
#Identify Months with Highest Sales
top_months <- monthly_revenue %>%
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5)

kable(top_months, caption = "Top 5 Months with Highest Sales")
```

Table 8: Top 5 Months with Highest Sales

YearMonth	TotalRevenue
2011-11-01	1563226
2011-10-01	1214040
2011-09-01	1100625
2010-12-01	868633
2011-05-01	813138

```r
#Visualise Monthly Sales Trend (Time-Series Plot)
ggplot(monthly_revenue, aes(x = YearMonth, y = TotalRevenue)) +
  geom_line(color = "steelblue", size = 1) +
  geom_point(color = "red", size = 2) +
  labs(
    title = "Monthly Revenue Trend",
    x = "Month",
    y = "Total Revenue"
  ) +
  theme_minimal()
```




```r
#Product Performance Over Time
monthly_product_sales <- data %>%
  group_by(YearMonth, Description) %>%
  summarise(
    MonthlyRevenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  )

#Identify Lowest-Selling Products Each Month
lowest_products <- monthly_product_sales %>%
  group_by(YearMonth) %>%
  slice_min(MonthlyRevenue, n = 1, with_ties = FALSE)

kable(lowest_products, caption = "Lowest-Selling Product Each Month")
```

Table 9: Lowest-Selling Product Each Month

YearMonth	Description	MonthlyRevenue
2010-12-01	LUNCH BOX WITH CUTLERY FAIRY CAKES	0
2011-01-01	OLD DOC RUSSEL METAL SIGN	0
2011-02-01	DOORMAT MERRY CHRISTMAS RED	0
2011-03-01	DIAMANTE BOW BROOCH RED COLOUR	0
2011-04-01	ACRYLIC HANGING JEWEL,BLUE	0
2011-05-01	ANTIQUE GLASS HEART DECORATION	0

YearMonth	Description	MonthlyRevenue
2011-06-01	ANTIQUE GLASS PLACE SETTING	0
2011-07-01	?	0
2011-08-01	STRAWBERRY DREAM CHILDS UMBRELLA	0
2011-09-01	CREAM AND PINK FLOWERS PONY	0
2011-10-01	ACRYLIC JEWEL SNOWFLAKE, PINK	0
2011-11-01	BLUE GIANT GARDEN THERMOMETER	0
2011-12-01	BLUE GIANT GARDEN THERMOMETER	0

```r
#Products Consistently Among Lowest-Selling
consistent_low_products <- lowest_products %>%
  group_by(Description) %>%
  summarise(
    MonthsAsLowest = n()
  ) %>%
  arrange(desc(MonthsAsLowest))

kable(consistent_low_products, caption = "Products Consistently Among Lowest Sellers")
```

Table 10: Products Consistently Among Lowest Sellers

Description	MonthsAsLowest
BLUE GIANT GARDEN THERMOMETER	2
?	1
ACRYLIC HANGING JEWEL,BLUE	1
ACRYLIC JEWEL SNOWFLAKE, PINK	1
ANTIQUE GLASS HEART DECORATION	1
ANTIQUE GLASS PLACE SETTING	1
CREAM AND PINK FLOWERS PONY	1
DIAMANTE BOW BROOCH RED COLOUR	1
DOORMAT MERRY CHRISTMAS RED	1
LUNCH BOX WITH CUTLERY FAIRY CAKES	1
OLD DOC RUSSEL METAL SIGN	1
STRAWBERRY DREAM CHILDS UMBRELLA	1

```r
#Identify Highest Sales Months
top_months <- monthly_revenue %>%
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5)

kable(top_months, caption = "Months with Highest Sales")
```

Table 11: Months with Highest Sales

YearMonth	TotalRevenue
2011-11-01	1563226
2011-10-01	1214040
2011-09-01	1100625
2010-12-01	868633

YearMonth	TotalRevenue
2011-05-01	813138

```r
# Identify top 5 highest revenue months
top_months <- monthly_revenue %>%
  arrange(desc(TotalRevenue)) %>%
  slice_head(n = 5)

# Display
kable(top_months, caption = "Months with Highest Sales")
```

Table 12: Months with Highest Sales

YearMonth	TotalRevenue
2011-11-01	1563226
2011-10-01	1214040
2011-09-01	1100625
2010-12-01	868633
2011-05-01	813138

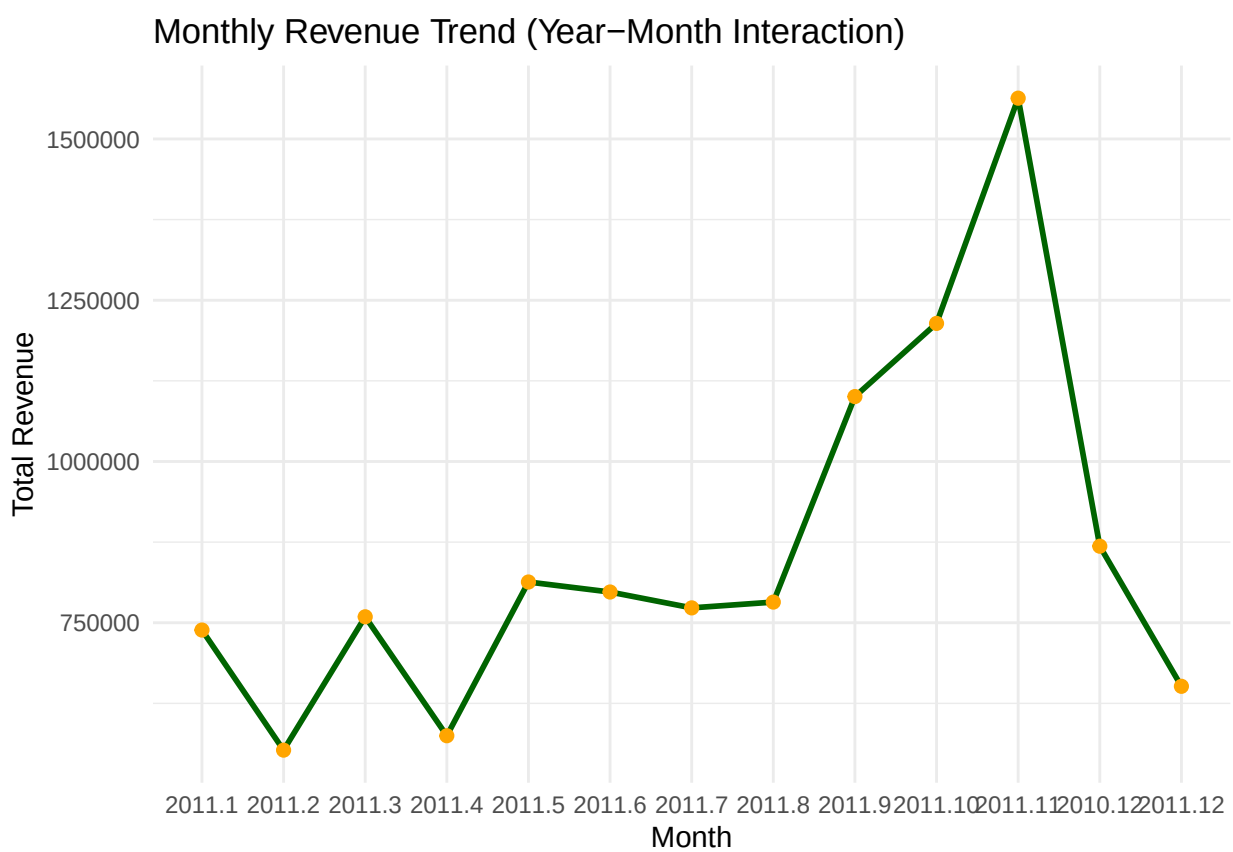
```r
# Calculate monthly revenue per product
monthly_product_sales <- data %>%
  group_by(Year, Month, Description) %>%
  summarise(
    MonthlyRevenue = sum(Revenue, na.rm = TRUE),
    .groups = "drop"
  )

# Identify lowest-selling product per month
lowest_products <- monthly_product_sales %>%
  group_by(Year, Month) %>%
  slice_min(MonthlyRevenue, n = 1, with_ties = FALSE)

# Display
kable(lowest_products, caption = "Lowest Selling Products per Month")
```

Table 13: Lowest Selling Products per Month

Year	Month	Description	MonthlyRevenue
2010	12	LUNCH BOX WITH CUTLERY FAIRY CAKES	0
2011	1	OLD DOC RUSSEL METAL SIGN	0
2011	2	DOORMAT MERRY CHRISTMAS RED	0
2011	3	DIAMANTE BOW BROOCH RED COLOUR	0
2011	4	ACRYLIC HANGING JEWEL,BLUE	0
2011	5	ANTIQUE GLASS HEART DECORATION	0
2011	6	ANTIQUE GLASS PLACE SETTING	0
2011	7	?	0
2011	8	STRAWBERRY DREAM CHILDS UMBRELLA	0
2011	9	CREAM AND PINK FLOWERS PONY	0

Year	Month	Description	MonthlyRevenue
2011	10	ACRYLIC JEWEL SNOWFLAKE, PINK	0
2011	11	BLUE GIANT GARDEN THERMOMETER	0
2011	12	BLUE GIANT GARDEN THERMOMETER	0

```r
#Monthly revenue trend with Year-Month interaction
monthly_revenue_interaction <- data %>%
  group_by(Year, Month) %>%
  summarise(TotalRevenue = sum(Revenue, na.rm = TRUE),
            .groups = "drop")

ggplot(monthly_revenue_interaction,
       aes(x = interaction(Year, Month),
           y = TotalRevenue,
           group = 1)) +
  geom_line(color = "darkgreen", size = 1) +
  geom_point(color = "orange", size = 2) +
  labs(
    title = "Monthly Revenue Trend (Year-Month Interaction)",
    x = "Month",
    y = "Total Revenue"
  ) +
  theme_minimal()
```



Interpretation

The time-series analysis shows fluctuations in revenue for different months of the year. Some months show significantly higher revenue, possibly indicating a purchasing season for the products. In the past, retail businesses have shown increased sales in months such as November and December, as people buy gifts and seasonal products. Months with low consumer demand are those with low revenue in the analysis. The analysis has also shown products with consistently low sales in different months.

Some of the characteristics of these products could be: -Low consumer demand for the products -Lack of product visibility -Outdated designs -High price compared to similar products

As a manager, the analysis can help in: -Improving inventory management -Changing the marketing approach -Launching promotions for products with low sales -Removing products with consistently low sales

The revenue trajectory also implies that there may be some seasonal effects on sales, as some months show high demand for products. The data may be used by businesses to:

-increase stock levels before high-demand months -create advertising campaigns during low-demand periods -manage staffing and logistics during high-demand periods